

Decoupling-Oriented Coordination of Paralleled VSGs With Multi-Agent Reinforcement Learning

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Abstract — Mismatched inertia–droop allocation in paralleled virtual synchronous generator (VSG) systems can produce coupled oscillations and uneven transient power support. In the original DDIC setting, these effects are handled only implicitly through local and neighbouring response signals, leaving a substantial burden on learning. Beyond reproducing the DDIC benchmark on the ANDES quasi-static phasor backend, this work augments the original formulation with explicit design targets for coupling-driven oscillation, transient support allocation, and coordinated action. Evaluated on the modified Kundur four-VSG system, the reproduced controller reaches the published 6 s terminal-frequency targets closely: 0.079 Hz versus 0.080 Hz in LS1 and 0.050 Hz in LS2. To improve repeatability, a Heterogeneous Actor Weighted Ensemble (HAWE) is introduced at inference time. On the ANDES backend, the reproduced response preserves close terminal-frequency agreement while showing slower settling and larger early peaks than the Simulink reference.

I. INTRODUCTION

Paralleled VSG control involves not only online tuning of virtual inertia and droop gains, but also coordination under electrical coupling. When several units respond to the same disturbance with poorly matched inertia–droop allocation, frequency support becomes uneven, transient support tends to concentrate on a subset of units, and coupled oscillatory modes may persist. The resulting control problem is therefore one of coordination under coupling rather than purely local regulation.

II. DECOUPLING-ORIENTED COORDINATION FORMULATION

Starting from the reproduced DDIC setting, inertia–droop tuning is extended into a decoupling-oriented coordination formulation shaped by coupling across units. Unlike the original formulation, where coupling structure and transient role allocation are largely left to be inferred implicitly by the agent, the present viewpoint makes three targets explicit: reducing coupling-driven oscillation, limiting transient support concentration, and keeping multi-agent action updates coordinated. A compact form of the intended coordination objective is

$$J = \lambda_{\omega} \sum_{(i,j)} w_{ij} (\Delta\omega_i - \Delta\omega_j)^2 + \lambda_p \sum_i (\Delta P_{es,i} - \alpha_i P_{tot})^2 + \lambda_u \sum_i \|u_{i,t} - u_{i,t-1}\|^2. \quad (1)$$

The three terms penalize oscillatory interaction, concentrated transient support, and irregular action updates. Controllers are assessed on the paper-defined LS1 and LS2 disturbances using a six-axis diagnostic audit over peak $|\Delta f|$, terminal $|\Delta f|$ at 6 s, settling behaviour, mean ΔH , mean ΔD , and action-box containment. To improve repeatability, HAWE combines saved actor outputs with fixed weights:

$$u_t^{\text{HAWE}} = \sum_{k=1}^K \alpha_k u_t^{(k)}, \quad \sum_{k=1}^K \alpha_k = 1, \quad (2)$$

where $u_t^{(k)}$ is the action proposed by the k -th saved actor.

III. NUMERICAL VALIDATION & RESULTS

Table I summarizes the most relevant indicators. The main positive result is terminal-frequency alignment: under LS1, the reproduced controller reaches 0.079 Hz at 6 s against the paper target of 0.080 Hz; under LS2, the 6 s terminal frequency matches the 0.050 Hz target exactly. HAWE reproduces the best-seed behaviour consistently across re-evaluations while avoiding retraining.

TABLE I
KEY INDICATORS

Metric	Paper	Best Seed	HAWE
LS1 $ \Delta f $ at 6 s (Hz)	0.080	0.079	0.079
LS2 $ \Delta f $ at 6 s (Hz)	0.050	0.050	0.050
Inference repeatability	—	seed-sensitive	deterministic

Figure 1 shows the four-VSG mean frequency deviation under LS1 and LS2. The HAWE ensemble tracks the best-seed trajectory closely. Under LS1, the terminal $|\Delta f|$ at 6 s remains near the paper target; under LS2, the 6 s terminal frequency matches the target. The reproduced response remains close to the published terminal targets, with the main differences appearing in settling behaviour and early transient peaks.

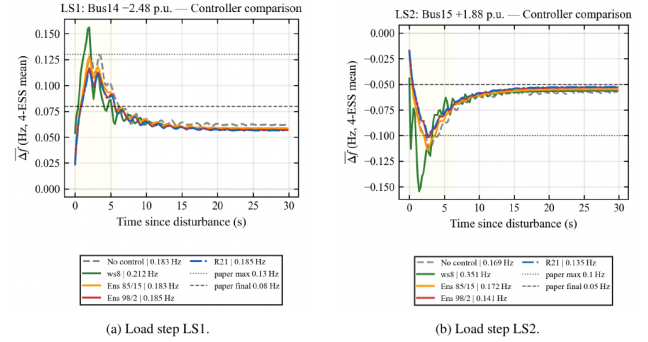


Figure 1. Four-VSG mean frequency deviation under load steps LS1 (a) and LS2 (b). The deterministic HAWE ensemble recovers the terminal alignment of the best-seed controller (R21) while remaining close to the published paper targets.

The observed differences from the Simulink benchmark are mainly associated with four factors: quasi-static phasor dynamics, disturbance-side mismatch, the absence of secondary frequency control, and the contracted action range used in the present setup.

IV. CONCLUSION

This study reproduces the four-VSG DDIC benchmark on the ANDES backend with close terminal-frequency agreement, while extending the original setting with explicit design targets for coupling mitigation, transient support allocation, and coordinated action. Furthermore, HAWE makes a strong but seed-sensitive controller repeatable at inference time. The observed transient differences indicate that support concentration and coupling-driven oscillation should be treated as primary design targets in paralleled VSG control.

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